AMANI KIRUGA

302-407-9899 ◇ Boston, MA
akiruga@fas.harvard.edu ◇ linkedin.com/in/amanikiruga ◇ amanikiruga.github.io

EDUCATION

PhD in Computer Science, Harvard University — Expected 2030
Advisor: Prof. Yilun Du | Research Focus: Scene understanding for vision and robotics

B.S in Computer Science, University of Delaware — August 2024
Cumulative GPA: 3.94 | Major GPA: 4.0

Selected Coursework: Machine Learning, Computer Vision, Calculus, Linear Algebra, Probability Theory, Computational Mathematics, Parallel Computing, Algorithms, Data Structures, Data Science

Bachelor Thesis: "Modelling 3D Causal Processes from 2D Vision Data: Challenges and Solutions"

SKILLS

Languages	Python, C++, C, Java, JavaScript, TypeScript, HTML, CSS
Frameworks/Libraries	Pytorch, OpenCV, NumPy, Pandas, Scikit-Learn, Node.js, React.js

EXPERIENCE

Student Researcher — Sep. 2023 - March 2025
MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) — *Cambridge, MA*

- Worked with Prof. Vincent Sitzmann and Prof. Ayush Tewari on unsupervised learning.
- Developed a method for out-of-domain generalization for single-image 3d reconstruction.

Student Researcher — June 2023 - Sep. 2023
MIT Media Lab — *Cambridge, MA*

- Worked with Dr. Andrew Lippman on content analysis in the Viral Communications Group.
- Designed and built an interactive portal to analyze media sources with LLMs using Next.js and MongoDB.

Student Researcher — Jan. 2024 - May 2024
University of Delaware, Dynamic Vision Lab — *Newark, DE*

- Worked with Prof. Christopher Rasmussen on video action recognition and pose detection.
- Benchmarked LLM performance, trained ResNet and DINO baselines using PyTorch.

Fullstack Software Engineer — Sep. 2021 - May 2024
University of Delaware, PAROT Group — *Newark, DE*

- Designed and built two releases of the National Science Foundation (NSF) funded Atom research portal that provides the scientific community with accessible atomic data.
- Utilized React, Node.js, Next.js, and PostgreSQL for robust application architecture.

Student Researcher — June 2021 - May 2023
University of Delaware, D-REAL Lab — *Newark, DE*

- Collected and annotated a human biomechanics dataset from YouTube with 1000 videos.
- Trained a video classification model with PyTorch to detect falls in an unconstrained environment.

PUBLICATIONS

- **UGG: Unified Generative Geometry Model** **A. Kiruga***, Z. Li*, R. Cai*, H. Lillemark, X. Wu, A. Tewari, Y. Du. NeurIPS 2026.
- **Infer3D: 3D Reconstruction through Inverse Generative Modelling** **A. Kiruga**, M. Hamilton, J.B. Tenenbaum, W.T. Freeman, V. Sitzmann, Y. Du, A. Tewari. Preprint 2026.
- **Learning Self-Orchestration in Decentralized Agent Swarms** G. Xu*, Z. Qi*, **A. Kiruga**, H. Lakkaraju, Y. Du. Preprint 2026.
- **A Portal for High-Precision Atomic Data and Computation A. Kiruga**, C. Cheung, P. Barakshan, A. Bhosale, R. Eigenmann, M. Safronova, B. Arora. Computer Physics Communications 2025.
- **Infant Video Interaction Recognition using Monocular Depth Estimation** C. Rasmussen, **A. Kiruga**, J. Orlando, M. Lobo. ISVC 2024 (Oral).
- **Non-Hierarchical Transformers for Pedestrian Segmentation A. Kiruga**, X. Peng. CVPR Workshop (AVA) 2023.

*Equal contribution.

PRESENTATIONS

- **Infant Video Interaction Recognition using Monocular Depth Estimation** C. Rasmussen, **A. Kiruga**, J. Orlando, M. Lobo. Poster, APTA CSM 2024.
- **Human Falls in The Wild: A Video Dataset** Z. Jia, **A. Kiruga**, X. Peng. Poster, UD Undergraduate Research Summer Symposium 2022.
- **Near Real-time Deep Group Profiling in Crowded Scenes A. Kiruga**, M. Baksh, K. Barner. Poster, UD Undergraduate Research Summer Symposium 2021.

PROJECTS

- **EmojiGAN** (Jan. 2022) Used PyTorch to train a Deep Convolutional Generative Adversarial Network (DCGAN) on an open-source emoji dataset of 4000+ pictures. Tested for GAN memorization using K-Nearest Neighbor search.

HONORS & AWARDS

- **MIT Summer Research Program Scholar** (June 2023 - Sep. 2023)
- **Google CS Research Mentorship Program Scholar** (Jan. 2022 - April 2022)
- **Eugene du Pont Memorial Scholarship** (2020-2024) - Full-ride scholarship awarded to 1 of 30,000 applicants based on high academic achievement, service, and leadership.
- **Outstanding Senior Student Award** (2024) - Awarded to 1 of 600 senior undergraduates in the CIS department.
- **Outstanding Sophomore Student/Paul D. Amer Meritorious Award** (2022) - Awarded to 2 of 600 sophomore undergraduates in the CIS department.
- **Rees Carl J & Eleanor K Mathematics Scholarship** (2022) - Awarded to 10 of 200 undergraduates based on high academic achievement in Mathematical Sciences.
- **Dean's List** (2021, 2022, 2023, 2024)

LEADERSHIP & SERVICE

Founder Feb. 2022 - Sep. 2024
University of Delaware, Competitive Programming Club *Newark, DE*

- Founded the UD Competitive programming club on campus and grew club to 100+ members.
- Organized weekly presentations and competitions to spread competitive programming knowledge.
- Received bronze medal in the Mid-Atlantic regional 2021 International Collegiate Programming Contest.

Teaching Assistant Sep. 2021 - May 2023
Microsoft Philanthropies TEALS, Trenton Central High School *Trenton, NJ*

- Assisted in the running of a class of 16 students in an AP Computer Science A classroom.
- Supported students to learn content related to Java by holding one-on-one help sessions and giving presentations.